

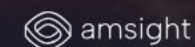
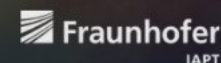
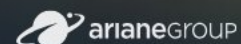
Materials built for the extreme.

Designed with AI. Made and tested in Europe.

FUNDED BY THE EUROPEAN SPACE AGENCY

PRISM's first deployment, through ESA's programme for future launchers (FLPP).

PRISM ALPHA, WITH



Vulcan 2 on test stand P5, DLR Lampoldshausen. Photo: DLR, CC BY 3.0.

THE PROBLEM

Far too many possible alloys to make them all.

474,311,376

possible alloys from five of nine high-melting metals, mixed in 1% steps

130,000 years

to make each one once, at ten a day

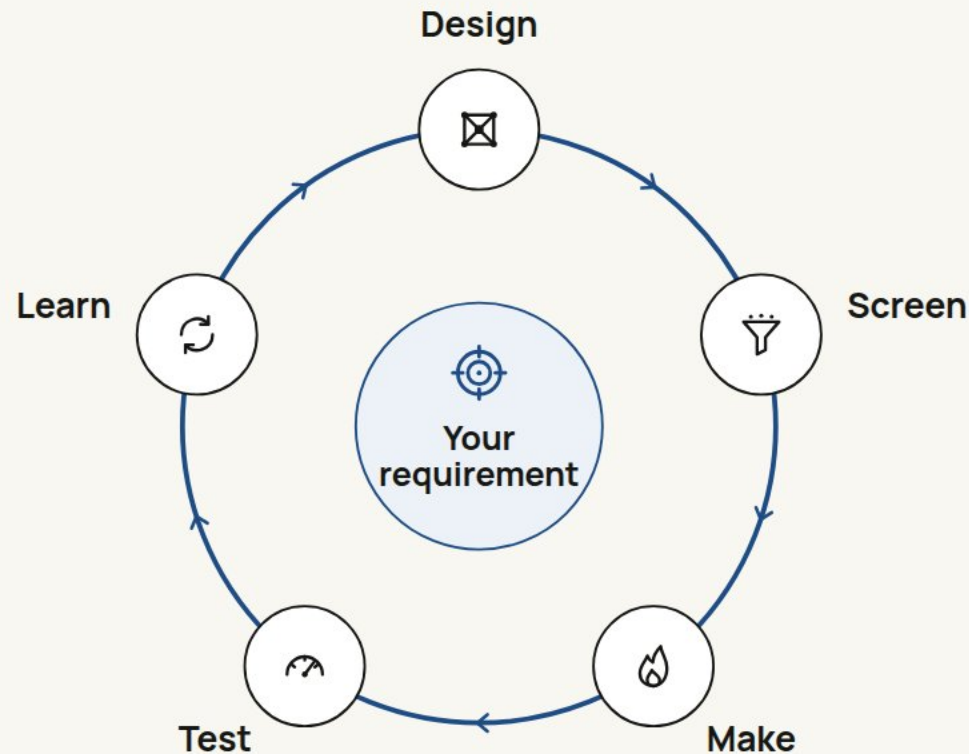
10–20 years

for a new material to reach service today

Sources: Arithmetic: $126 \text{ ways to pick five of nine metals that melt above } 1,650^\circ\text{C} \times 3,764,376 \text{ mixes in whole percent}$. The field is to scale; its layout is illustrative.



One loop: design, make, test, learn.

**Requirement**

You tell us what the part must survive.

PROTOTYPE

**Design**

AI suggests new mixes, from the whole range.

PROTOTYPE

**Screen**

Simulations rule most out, before anything is melted.

PROTOTYPE

**Make**

We melt the best ideas, make powder and print them.

IN USE

**Test**

Each sample is measured against your targets.

IN USE

**Learn**

Every result goes back into the models.

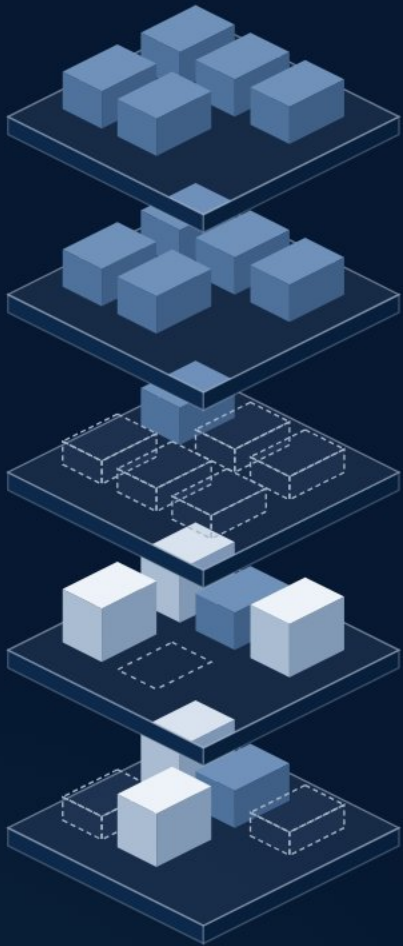
PROTOTYPE

FIRST TARGET

Refractory high-entropy alloys for oxygen-rich preburners in rocket engines, to replace Monel K500 and other legacy alloys.

Five stacks, one system.

■ In use ■ Prototype □ In development --- Goal



01 Research

AI suggests mixes; physics simulations throw out what cannot work.

PROTOTYPE

02 Harness

Plans each round, runs the tools, scores the results, remembers failures.

PROTOTYPE

03 Autonomy

Robots weigh, melt and measure. People stay in charge.

IN DEVELOPMENT

04 Manufacturing and test

Melting, powder, laser printing and testing, with Bimo Tech and Fraunhofer IAPT.

IN USE

05 Evidence

Every result keeps its source, its owner and its rules.

IN USE

Maturity: Each block on a plate is one component, as tall as it is built. In use: runs in current programmes. Prototype: PRISM software, being taken from TRL 3 to 4 in PRISM Alpha. In development: being built.

Four modules, one loop.



Loop: The planner sends a campaign; the samplers propose candidates; simulation and the lab test them; results are kept, with sources; what worked and what failed plans the next round. An engineer signs off what leaves.

Physical synthesis supplies the reward signal the models train on.

01 Recipe translation

PRISM outputs target recipes: precursors, temperatures and thermal profiles.

02 Robotic synthesis

Robot arms dose powder and move samples through heating profiles.

03 Automated characterisation

Diffraction patterns are captured from every synthesised sample.

04 Phase identification

Neural networks compare patterns against structure databases to confirm phases.

05 Feedback

Structured results return to the Evolver's reflector and change the next campaign.

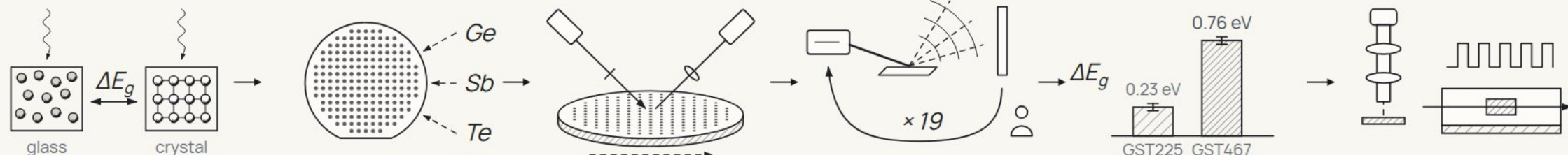
Maturity: In development. Direct instrument control closes the loop in phase III of the plan.



The melt spinner in the materials lab at WUST, Wrocław, where our alloys are melted. We rent the equipment.

Every result carries its own proof.

A REAL LINEAGE, PUBLISHED BY NIST IN 2020



01 · THE QUESTION

Which Ge-Sb-Te mix changes most between glass and crystal?

○ PUBLIC

02 · THE OPTIONS

177 mixes, side by side on one wafer.

○ PUBLIC

03 · WHAT WAS KNOWN

A light scan of every spot, before the run.

○ PUBLIC

04 · THE MEASUREMENTS

19 rounds. Each time, the AI picked what to X-ray.

○ PUBLIC

05 · THE ANSWER

$\text{Ge}_4\text{Sb}_6\text{Te}_7$: about three times the contrast of GST225.

○ PUBLIC

06 · THE PROOF

Confirmed under an electron microscope and in a working device.

○ PUBLIC

CAMEO, 19 rounds

about 10 hours



Measuring all 177

more than 90 hours



RIGHTS TRAVEL WITH EVERY RECORD

Published data ○ PUBLIC

Partner data ■ PRIVATE

Result ■ PRIVATE

A result keeps the strictest rights of its inputs.

FOUR LEVELS OF SHARING

■ PRIVATE

Stays with its owner.

○ RELEASED

Shared with named partners.

☑ COMPUTABLE

Software may use it. No person reads it.

○ PUBLIC

Published on purpose.

WHAT WE PROTECT

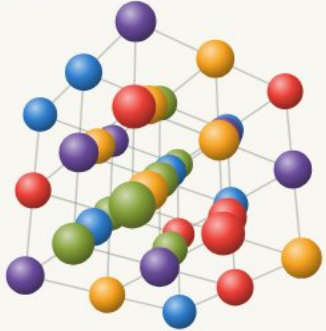
Generative sampling, conditioned on thermodynamic feasibility, reaches metastable alloys that equilibrium screening excludes.

PATENT FILINGS IN PROGRESS

Sources: CAMEO is the work of NIST and its partners, not ours: Kusne et al., Nature Communications 11, 5966 (2020). PRISM keeps the same record for every result: provenance and export labels on ESA work today, automatic enforcement of sharing rules in development.

Funded, contracted and in the lab.

ACTIVE



Project SPARK

ESA activity · Bimo Tech

8

Refractory high-entropy alloy candidates, taken to two physical down-selections.

RUNNING



PRISM Alpha

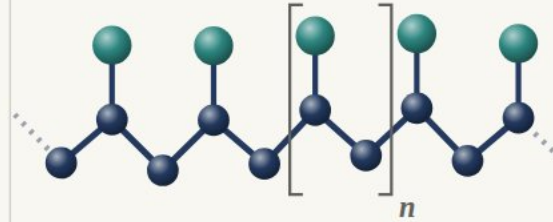
ESA · future launchers programme (FLPP)

TRL 3 → 4

In 12 months: at least three candidates and one complete closed loop.

With arianeGROUP Fraunhofer IPT amsight

CONTRACTED



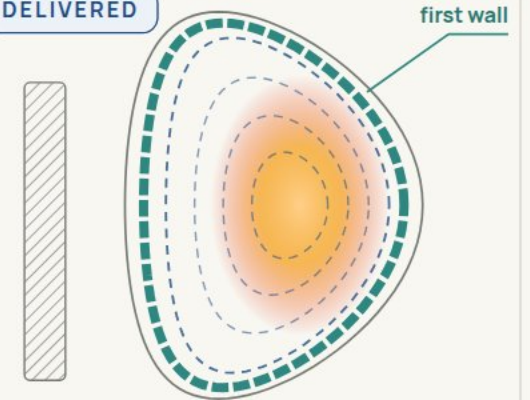
PFAS-free polymers

Industrial partner · under NDA

1st

Privately funded programme, and our first polymer class. Contract and NDA signed; start pending.

DELIVERED



Fusion heritage

Bimo Tech · ITER

ITER

Titanium first-wall materials and rhodium targets, supplied by Bimo Tech.

Sources: ESA contract records; Mirdyne and Bimo Tech programme records.

Where PRISM goes first.

01 · NOW



Space propulsion

Alloys for the hottest parts of rocket engines. First with ESA and ArianeGroup.

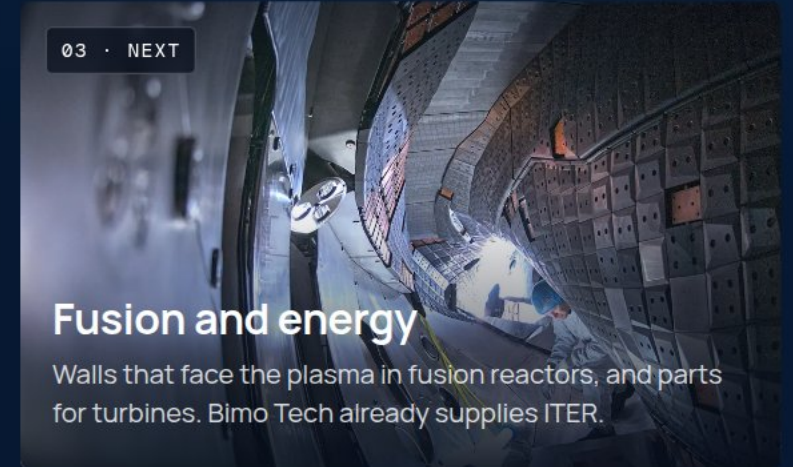
02 · NEXT



Defence

Hot-section materials for aircraft engines, with export rules built in.

03 · NEXT



Fusion and energy

Walls that face the plasma in fusion reactors, and parts for turbines. Bimo Tech already supplies ITER.

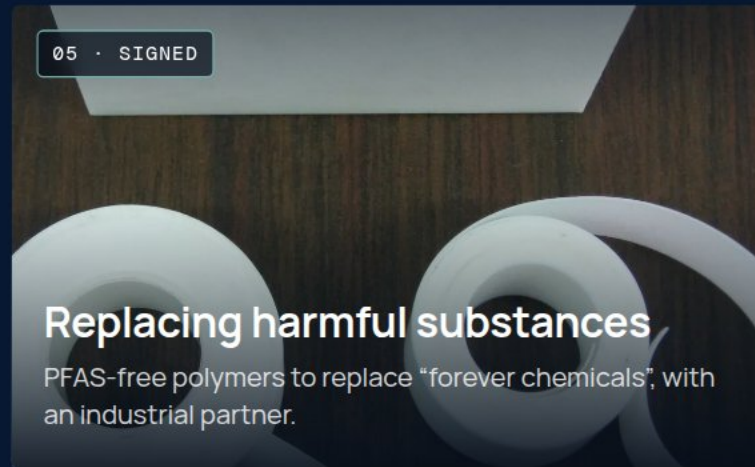
04 · NEXT



Supply independence

Alloys that do the job with metals Europe can get.

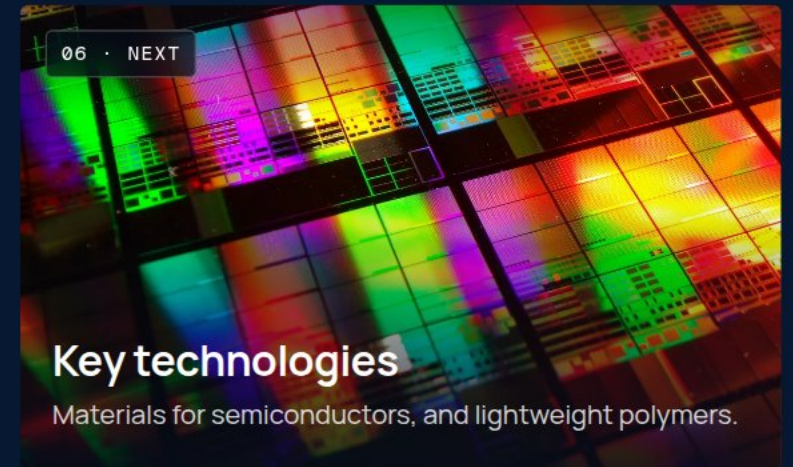
05 · SIGNED



Replacing harmful substances

PFAS-free polymers to replace "forever chemicals", with an industrial partner.

06 · NEXT

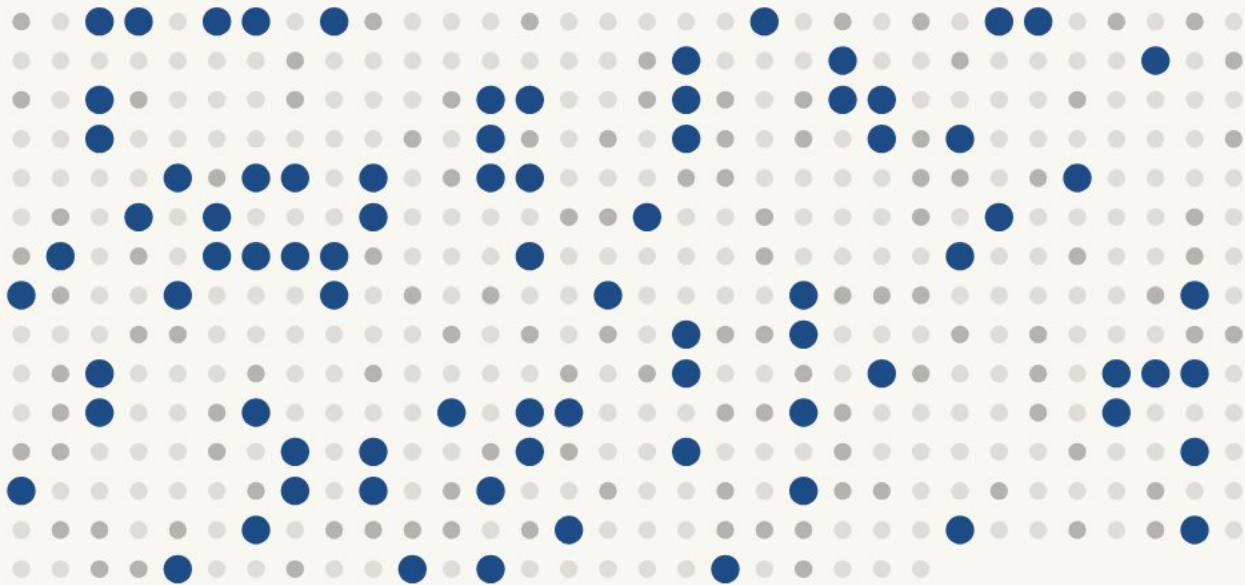


Key technologies

Materials for semiconductors, and lightweight polymers.

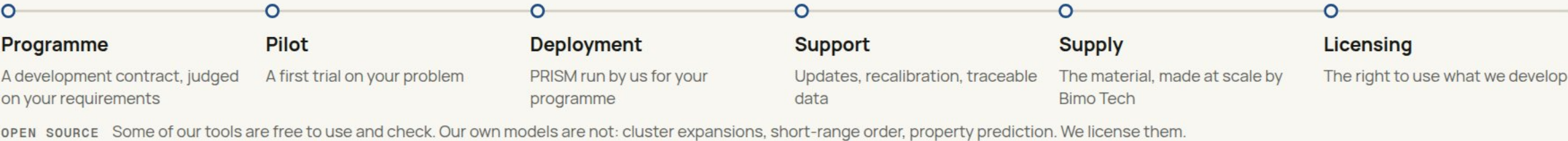
Photos: Aestus engine in test stand P4.2: DLR, CC BY 3.0 · EJ200 afterburner: Julian Herzog, CC BY 4.0 · Wendelstein 7-X wall tiles: Christopher Roux, EUROfusion, CC BY 4.0 · Tungsten crystals: Alchemist-hp (pse-mendelejew.de), Free Art License. All cropped.

80 companies we can reach spend €48M a year.



472	aerospace companies in the EU with 20 or more staff
189	work on what we make: materials, propulsion, qualification
80	we can reach through our network, 2026 to 2031
€48M	a year: what these 80 spend on developing and qualifying materials, about €600k each

HOW WE EARN



Sources: Companies: EU27, 2024 (provisional), from the evidence register of the PRISM financial model. Fit, reach and spend per company are the model's assumptions (research refresh, 27 Jul 2026).

The money is moving into AI that makes materials.



PRISM, IN EUROPE

PRISM
The idea

PRISM to ESA
Submitted to OSIP

PRISM Alpha
ESA project in FLPP

First private project
PFAS-free polymers

THE MONEY, IN THE US

2024

JUL 2025

JUL 2025

SEP 2025

OCT 2025

NOV 2025

JUL 2026

AUG 2026

AUG 2026

Radical AI \$55M
Seed round

Periodic Labs \$300M
Seed round

Lila Sciences \$550M
Raised in total

Genesis Mission
US government programme

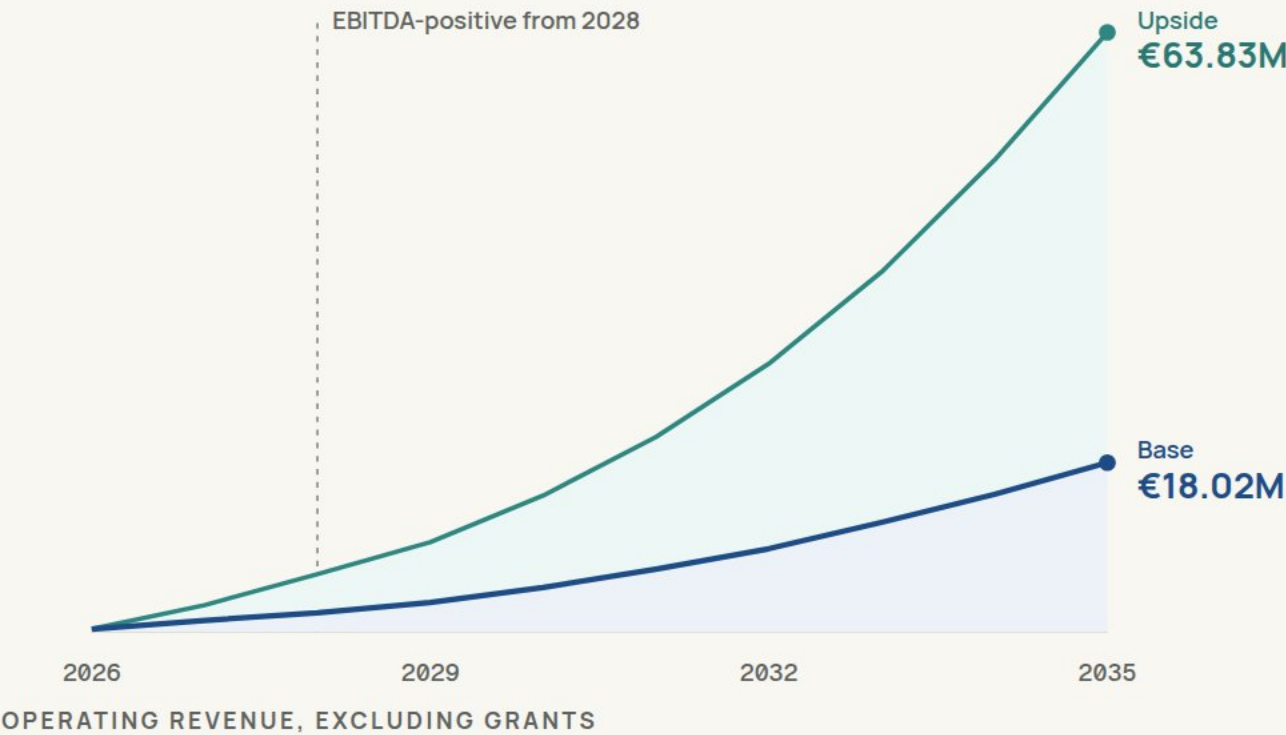
Discovery Loop
Hundreds of millions raised

Almost all this money is in the US. PRISM designs, makes and tests in Europe.

Earth: [EOxCloudless](#) by EOx IT Services GmbH (contains modified Copernicus Sentinel data 2016), [CC BY 4.0](#)

Sources: Company announcements: Radical AI, July 2025; Periodic Labs, 30 September 2025; Lila Sciences, October 2025 (seed and Series A in total). The White House, "Launching the Genesis Mission", 24 November 2025. Discovery Loop: TechCrunch, 5 August 2026.

Ten-year operating plan.



61%
gross margin in 2035
Upside 81%

2028

the first year with a positive EBITDA
Upside 2027

€2.9M

equity needed in total
Upside €0.6M

Growth is set by the hiring plan: programmes count as revenue only up to the delivery capacity left after pilots, deployments and support.

Source: PRISM financial model, research refresh 27 Jul 2026 · checks pass · € nominal. EBITDA 2035, before grants: €4.71M base, €40.84M upside.

The team that runs PRISM's ESA work.

KG

Kevin Grüning

CEO AND MANAGING DIRECTOR,
MIRDYNE

Space Systems Lead at Bimo Tech. Physics and technology for space applications, JLU Giessen and THM.

SK

Siddhartha Yash Kovid

TECHNICAL LEAD, MIRDYNE

Technical lead of the ESA projects SPARK and PRISM Alpha at Bimo Tech. Applied AI and data science, MIT Professional Education; biomedical engineering, THM.

MO

Marcin Orzechowski

CFO, MIRDYNE

CEO and Head of R&D at Bimo Tech, which supplies special metals and precision parts for space, energy and science. Wrocław University of Technology.



KI-Sonderpreis, Hessen Ideen 2026

The special prize for artificial intelligence. Photo: Hessen Ideen.



Mirdyne is a spin-off of Bimo Tech, which makes special metals and precision parts and supplies ITER.

ADVISORS

Prof. Jan Wróbel, Warsaw University of Technology · Prof. Dariusz Jarząbek, IPPT PAN

€4M

Seed round, closing 15 December 2026.

The plan needs €2.91M never to run out of cash. We are raising €4M so the plan survives being wrong, and so the downside gate closes before the money does.

WHERE THE MONEY GOES



- **Team** 65%
Delivery, materials science, product and security headcount
- **Go-to-market, IP and export control** 16%
Bids, consortia, field selling, patents, dual-use compliance
- **Compute, infrastructure and lab access** 13%
Controlled core, per-site secure compute, partner-lab access
- **Probe hardware, tools and testing** 6%
Probe builds, calibration, R&D tooling

ROADMAP · 24 MONTHS

NOW

6 MONTHS

Computational validation

The full discovery loop running the first PRISM campaigns.

12 MONTHS

Hybrid loop

Integration with Fraunhofer IAPT additive-manufacturing data.

24 MONTHS

Autonomous laboratory

Direct instrument control closes the loop with no human in the sequence.

Start with the capability you need.

Alloys and polymers, designed, made and tested in one loop, with the proof attached.

Register interest →

mirdyne.com →

The European Space Agency funds PRISM's first deployment, through its programme for future launchers (FLPP).

